

SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	13300	Calculated Profit Factor:	11.28
Win Accuracy:	82.6% (10984W / 2316L)	Max Peak Drawdown:	12.9%
Cumulative PnL:	\$19844.80	Avg. PnL per Trade:	\$1.49

EQUITY CURVE PROGRESSION



PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-2660	35.7%	\$-897.27	+29% [OK]
Trades 2661-5320	92.1%	\$2966.93	+111% [OK]
Trades 5321-7980	100.0%	\$5149.64	+120% [OK]
Trades 7981-10640	100.0%	\$8760.32	+120% [OK]
Trades 10641-13300	85.2%	\$3865.18	+102% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SOVEREIGN AI: SYSTEMIC RISK & ALGORITHMIC AUDIT REPORT

Target System: Multi-Asset Stochastic Mean-Reversion Engine

Dataset Scope: 13,300 Settled Trades

Status: CRITICAL RISK ALERT (Asymmetric Tail-Risk Vulnerability)

1. Executive Telemetry Critique

The high-level performance metrics present a highly anomalous and dangerous divergence:

$$\text{Win Rate} = 82.6\% \quad | \quad \text{Profit Factor} = 11.28 \quad | \quad \text{Average PnL/Trade} \approx \$1.49$$

The Synthetic Win-Rate Illusion: A win rate of 82.6% combined with a Profit Factor of 11.28 on 13,300 trades typically indicates an institutional-grade alpha engine. However, the average PnL per trade of \$1.49 exposes severe structural decay. Transaction costs, slippage, and borrow fees in any live execution environment will instantly render this system net-negative.

The Fatal Multiplier Overflow: The parameter `atrStopMultiplier` is currently set to `9.219838168703331e+30`. This is a floating-point system overflow. The system effectively has no stop-loss.

De Facto Risk Profile: Because the ATR stop is mathematically infinite, the system never exits a losing trade via risk-based stop-outs. Instead, it relies entirely on:

1. Mean reversion occurring before the time-out.
2. The time-based hard exit (`maxTicksInTrade: 120`).

Drawdown Underestimation: The maximum peak drawdown of 12.9% is highly deceptive. It is a historical artifact of a benign regime. Under a sustained, non-mean-reverting trend (e.g., a liquidity black hole or momentum breakout), the lack of a functional ATR stop-loss will result in 100% capital liquidation.

2. Regime Suitability & Behavioral Patterns

[Market Regime: Mean-Reverting / Range] ---> High Win Rate (82.6%) / Low PnL per Trade

[Market Regime: Strong Trend / Breakout] ---> Catastrophic Tail Risk (Infinite ATR Stop)

Current Suitability: This configuration is strictly limited to low-volatility, highly mean-reverting micro-ranges. The engine capitalizes on minor noise fluctuations.

Behavioral Pathology:

Loss-Nursing: The algorithm holds toxic, underwater positions for up to 120 ticks, hoping for a reversion to the mean. It "nurses" losses to maintain an artificially inflated win-rate metric.

Confluence Inefficiency: With `minConfluenceScore: 3` and a high Bollinger Band standard deviation of `2.25`, entries are highly selective. However, once entered, the risk management is non-existent due to the broken ATR parameter.

ADX Paradox: The `regimeAdxThreshold: 25` is designed to filter out strong trends. However, if ADX spikes after entry, the engine is trapped in a trending market with no stop-loss for 120 ticks.

3. Calibrated Recommendations & Parameter Boundaries

To transition this system from a high-risk liability to an institutionally viable asset, the parameters must be re-calibrated immediately to enforce strict risk boundaries.

```
```json
{
 "rsiOversoldThreshold": 30,
 "rsiOverboughtThreshold": 70,
 "bbPeriod": 20,
 "bbStd": 2.00,
 "maxTicksInTrade": 80,
 "minConfluenceScore": 3,
 "atrStopMultiplier": 2.50,
 "regimeAdxThreshold": 20
}
```

#### # Parameter Justification:

`atrStopMultiplier` (9.22e+30  $\rightarrow$  2.50): Critical Action. Re-establishes a hard, volatility-adjusted stop-loss. A 2.5x ATR multiplier allows sufficient breathing room for noise while clipping tail-risk. Win rate will decrease to an estimated 62–68%, but the system will become survivable.

`rsiOversold/Overbought` (25/75  $\rightarrow$  30/70): Widening the boundaries increases execution frequency, helping to counteract the low average PnL per trade by capturing more standard mean-reversion setups.

`bbStd` (2.25  $\rightarrow$  2.00): Lowering the Standard Deviation threshold to 2.00 increases trade frequency, allowing the system to scale its volume to offset the corrected, lower win rate.

`maxTicksInTrade` (120  $\rightarrow$  80): Reduces capital lockup and time-exposure in toxic, non-reverting trades.

`regimeAdxThreshold` (25  $\rightarrow$  20): Tightens the trend filter. By blocking entries when  $ADX > 20$ , the system avoids entering mean-reversion trades during nascent trend formations.

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## SOVEREIGN RISKS SENTINEL SYSTEMS

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